

RHEA SUDHEER

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SUMMARY

Software engineer with an M.S. in Computer Science from Texas A&M and hands-on experience building backend applications, concurrent systems, low-level software, and real-time ML-enabled services. Strong in C++, Python, Java, SQL/MySQL, data structures & algorithms, object-oriented design, Git, debugging, distributed-systems concepts, and SDLC. Experienced designing modular components, diagnosing reliability failures, testing and validating system behavior, and translating ambiguous requirements into maintainable software.

EDUCATION

Texas A&M University — Master of Computer Science, GPA 3.83/4.00 | College Station, TX Aug 2024 - May 2026

PES University — B.Tech. Computer Science, GPA 9.1/10 | Bangalore, India Dec 2020 - May 2024

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Analysis of Algorithms, Information Storage & Retrieval, Computer / Network Security

TECHNICAL SKILLS

Languages: C++, Python, Java, C, SQL, PHP, Bash

Core Software Engineering: Data Structures & Algorithms, Object-Oriented Design, SDLC, REST APIs, Unit Testing, Code Review, Debugging, System Reliability

Systems: Linux, Multithreading, Synchronization, Virtual Memory, Paging, Scheduling, Thread-Safe Queues, Backpressure, Distributed-Systems Concepts

Data & Backend: MySQL, Relational Schemas, ETL/Data Pipelines, Data Validation, Spark, Kafka, Flask, Laravel

Tools & Platforms: Git, CI/CD, Docker, Kubernetes, AWS, GCP, Jenkins, QEMU/Bochs, GDB/LLDB, CMake

EXPERIENCE

Public Policy Research Institute, Texas A&M University — Student Research Assistant Jan 2025 - Present

- Engineered and maintained a PHP/Laravel + MySQL platform supporting 100,000+ public records; designed backend services, relational schemas, REST/API integrations, internal tools, dashboards, and data-collection workflows.
- Built reusable SQL and ETL components for ingesting, joining, transforming, and validating multi-source data; automated checks for malformed, duplicated, incomplete, and inconsistent records before downstream use.
- Debugged application, database, and data-quality failures using logs, SQL investigation, targeted tests, and profiling; converted recurring failure modes into fixes, stronger validation, and regression protection.
- Improved maintainability through unit testing, CI checks, code review, Git-based development, documentation, and release support; partnered with researchers and developers to translate evolving requirements into technical specifications and iterative releases.
- Owned work across backend engineering, database design, ETL/data-quality logic, debugging, and release support; prioritized competing requests and kept application behavior, schema changes, and stakeholder-facing outputs synchronized as requirements evolved.

Thales — Artificial Intelligence Intern | Bangalore, India Jun 2022 - Aug 2022

- Developed a real-time MobileNetV2 + SSD pedestrian-detection pipeline; profiled latency, detection quality, missed/false detections, and model-size trade-offs under deployment constraints, and used SHAP/LIME/saliency analysis to debug and validate behavior.
- Balanced speed, interpretability, and failure analysis in a safety-sensitive perception workflow, documenting experiments and incorporating feedback to improve reliability and usability under real-time constraints.

Ernst & Young — Technology Consulting Intern | Bangalore, India Jan 2024 - Jun 2024

- Evaluated enterprise Oracle/SAP systems for access, change-management, disaster-recovery, security, and operational reliability; investigated exceptions with client technology teams and translated root causes into traceable remediation actions.

IBM — Tech Mentee | Bangalore, India Dec 2022 - Mar 2023

- Built a reproducible Python/Scikit-learn ML pipeline for Type-I diabetes prediction from heterogeneous multi-omic data, covering validation, preprocessing, feature engineering, model selection, cross-validation, and inference; achieved 97.7% reported accuracy.
- Compared decision-tree and ensemble approaches through controlled evaluation and presented methodology, validation results, and technical trade-offs to technical and non-technical audiences, incorporating mentor feedback into iterative improvements.

PROJECTS

Custom x86 Operating System | C/C++, x86, QEMU/Bochs, GDB/LLDB Systems Programming

- Implemented demand-paged virtual memory with page tables, page-fault handling, physical-frame allocation, and TLB-aware state management; built timer-interrupt-driven preemptive scheduling with ready queues and safe thread lifecycle handling.
- Implemented spinlocks/mutexes, an inode-based file system, buffered I/O, and interrupt-driven non-blocking disk access using request queues; debugged memory, scheduling, synchronization, and device-I/O failures at the hardware/software boundary.
- Separated scheduling state from thread execution and coordinated virtual-memory, file-system, synchronization, and device-I/O invariants to preserve correctness under asynchronous preemption.

Adaptive Multithreaded Job Scheduler with Backpressure | C++, Multithreading Oct 2025

- Built a concurrent scheduler with FIFO and priority policies, thread-safe queues, mutexes, condition variables, and explicit locking boundaries; added backpressure plus queue-depth, wait-time, and worker-utilization metrics to analyze overload and bottlenecks.
- Used queue depth, task wait time, and worker utilization to distinguish scheduling overhead from insufficient worker capacity and queueing delay under load.

Vision for the Blind: Real-Time Assistive Computer Vision | Python, TensorFlow, Flask Dec 2023

- Built an end-to-end real-time service combining camera input, YOLOv3 obstacle detection, LSTM temporal modeling, Flask serving, and audio feedback; designed around inference latency, temporal consistency, and usefulness of human-facing output.

Hybrid Self-RAG with External Knowledge Validation | Python, FAISS, LangChain May 2025

- Built a modular multi-stage system combining generation, self-checking, claim extraction, live API evidence validation, semantic scoring, conditional FAISS retrieval, and evidence-conditioned regeneration; separated components so failures and latency costs could be isolated and debugged.
- Implemented threshold-driven fallback behavior to avoid unnecessary retrieval/regeneration, improving factual accuracy from 81% to 89% while reducing hallucination rate from 19% to 11%; analyzed latency and source-availability trade-offs for system reliability.

PUBLICATIONS

[Digital Twins in Healthcare](#) — IEEE Access 2024 | [Robust Audio-Visual Speech Recognition](#) — AMATHE 2024 | [Leukaemia: Comparative Deep Learning Analysis](#) — RAICS 2024